DHRUV SHAH

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PROFILE

Hardware Systems Engineer & M.Sc. Embedded Systems graduate with robust hands-on experience in power electronics, PCB design, and embedded firmware. Demonstrated history of taking hardware from concept to commercial prototype, including designing a multi-wavelength illumination platform, developing power management circuits (PMC) for energy harvesters, and optimizing electric vehicle controllers. Strong technical foundation in multilayer PCB layout, embedded C (SPI/I2C), and circuit simulation (LTSpice, MATLAB), focused on building efficient and reliable hardware architectures.

WORK EXPERIENCE

Master's Thesis Student & Product Development Intern — 11/2024 - 05/2026
Opto BIOLABS — *Freiburg, Germany*

- Designed a 5V direct-drive multilayer PCB (120x155mm) in Fusion 360, collaborating on mechanical enclosure integration and engineering an active load switch to safely route 10A peak currents.
- Built custom C firmware and an SPI library for the MCU, establishing reliable cross-platform control and managing hardware decoders to achieve true 16-bit PWM resolution.
- Automated a high-density 324-LED matrix layout via Python to accelerate the product design iteration cycle by 30%, establishing a highly reproducible field with just 4.62% spatial non-uniformity.
- Integrated an I2C-controlled active thermal management system, ensuring long-term reliability for end-user biological assays by sustaining high-power irradiance (>60 mW/cm^2) with only 0.61% temporal drift over 12-hour runs.

Technical Inspector (Part-Time) — 01/2026 - Present
Industrial QC — *Germany*

- Executed Factory Acceptance Tests (FAT/IFAT) and witnessed performance testing for industrial hardware across major manufacturing facilities, ensuring strict compliance with engineering standards.
- Conducted visual testing (VT) and dimensional validations for critical industrial equipment, including 33kV GIS switchgear, electric actuators, and precision process instrumentation at tier-1 facilities such as Siemens, Endress+Hauser, VEGA, and AUMA.
- Managed critical inspection documentation, including nonconformity reporting, and coordinated between manufacturers and end-clients to ensure quality requirements and delivery timelines were met.

Power Electronics Designer — 10/2023 - 01/2025
IMTEK, Uni Freiburg — *Freiburg, Germany*

- Designed and tested a Power Management Circuit (PMC) for an electret-based wind energy harvester, achieving a 15% efficiency increase over previous models during sample load testing.
- Modeled circuit behavior using LTSpice/PSpice to reduce design iterations by 30%, developing a DC-AC mimicking circuit that simulated harvester output and cut production testing costs by 70%.
- Optimized energy harvesting outputs to successfully power 3 distinct small-scale devices, concurrently developing smart distribution architectures to target a 10% performance improvement.

Hardware Developer — 09/2024 - 12/2024
Intelligent Embedded Systems — *Freiburg, Germany*

- Designed a 4-layer PCB in Altium Designer to interface proximity sensors with an I2C address changer, mitigating signal interference by 20% and ensuring robust multi-sensor communication.

Embedded System Developer 08/2021 - 09/2022
Techinnovate Mobility *India*

- Designed and optimized custom electric vehicle controllers, enhancing overall vehicle performance by 15% and control responsiveness by 20%.
- Integrated Battery Management Systems (BMS) with digital displays, improving monitoring accuracy by 25% and reducing component failure rates by 30%.

Electrical Head 03/2019 - 07/2021
Team Vyadh – VIT Vellore *Vellore, India*

- Directed a 12-member electrical team as a two-time finalist in the University Rover Challenge (2020, 2021), achieving 95% robotic arm control precision to increase task execution speed by 25%.
- *Project Media:* 2021 URC Demonstration (Featured Speaker) | 2020 URC Overview

EDUCATION

Master of Science in Embedded Systems Engineering 10/2022 - 05/2026
Albert-Ludwigs-Universität Freiburg, Germany
Focus Area: Circuits & Systems
Master's Thesis: Development and Characterisation of a Multi-Wavelength Large-Area Illumination Platform for Optogenetics (Completed at Opto BIOLABS)
Relevant Coursework: Power Electronics for E-Mobility, Hardware Security and Trust, Mobile Robotics

Bachelor of Technology in Electrical and Electronics Engineering,
Vellore Institute of Technology, Vellore, India 05/2018 - 05/2022

CERTIFICATIONS

Embedding Sensors and Motors Specialization 02/2026
University of Colorado Boulder (Coursera)
Focus Areas: Sensor Manufacturing & Process Control, Motor Control Circuits, Signal Conditioning

SKILLS

Hardware	Altium, Eagle, Fusion 360, PCB Layout (Multilayer), DFM/DFT
Embedded	C/C++, Python, Register-Level Programming, Driver Dev, SPI/I2C/UART
Concepts	Power Electronics, Signal Integrity, Thermal Management
Tools & Lab	MATLAB, LTSpice/PSpice, Oscilloscope, Logic Analyzer, FAT/IFAT Testing

PUBLICATIONS & AWARDS

- **Alumni Prize (July 2024):** Awarded by the Faculty of Engineering, University of Freiburg, recognizing exceptional community leadership as an ICM Mentor, inter cultural events host, and Founder's Club Board Member.
- **IEEE SusTech (2024):** Co-authored published conference paper: "On the Use of an Electret-Based Wind Energy Harvester to Power a Vibration Sensor."
- **IEEE NPEC:** First author of published conference paper: "Energy Optimization for Solar Powered Rural Agro Loads with Elevated Energy Storage System."